AHMED SHEHAB

Data Analyst | Business Intelligence Analyst | Data-Driven Decision Maker
Nozha, Cairo, Egypt | +20 102 336 6797 | ahmedshehabds2@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Results-oriented Data Analyst and Business Intelligence Analyst with hands-on experience in end-to-end data analytics, statistical modeling, and data visualization, including hands-on internships at Telecom Egypt (WE), Giza Systems / Pulse, and iSchool Egypt. Proficient in Python, SQL, Power BI, Excel, and R, with a proven ability to transform raw data into actionable insights that drive strategic business decisions. Experienced in building predictive models, NLP pipelines, RAG-based AI systems, MLOps pipelines, and automating data workflows using AI Automation (n8n), and communicating complex findings clearly to both technical and non-technical stakeholders. Pursuing a Bachelor's in Business Technology with a specialization in Business Analysis.

CORE COMPETENCIES & TECHNICAL SKILLS

Programming & Querying: Python, R, SQL, Pandas, NumPy, Scikit-learn
Visualization & BI: Power BI, Microsoft Excel, Matplotlib, Seaborn, Plotly, Dashboard Design, Data Storytelling, Streamlit
Analytics & Modeling: EDA, Predictive Modeling, Regression Analysis, Classification, Customer Segmentation, RFM Analysis, Statistical Hypothesis Testing
NLP & AI Systems: Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant), Transformer Model Fine-Tuning (BERT, AraBERT, MARBERTv2, mT5), OCR, Text Classification, Summarization, Sentiment Analysis
MLOps & Deployment: MLflow (experiment tracking, model registry), Docker & Docker Compose, Apache Airflow (DAG orchestration), Flask REST API Design, PostgreSQL, Optuna hyperparameter tuning, CI-style automated retraining pipelines, Async Job Queuing
LLM Systems & Evaluation: Prompt Engineering, LLM-as-Judge Evaluation Design, Structured JSON Output Schemas, Google Gemini API, Retrieval Relevance Scoring/Gating, Test-Set Design for RAG Evaluation
Data Management: Data Cleaning & Preprocessing, Feature Engineering, KPI Development, Data Wrangling, Descriptive & Predictive Analytics
Automation: AI Automation: n8n, Workflow Automation, Data Pipeline Automation
Soft Skills: Critical Thinking, Problem Solving, Business Requirements Analysis, Communication

PROFESSIONAL EXPERIENCE

Data Analyst Intern — Telecom Egypt (WE) *Jun 2026 – Jul 2026*

- Built an end-to-end subscriber plan-recommendation pipeline across 5 relational datasets (~10,000 subscribers), merging customer, plan, payment, and network tables and engineering features including tenure, average monthly revenue, addon ratio, and price-to-quota ratio.
- Defined business-rule-based Upgrade / Downgrade / Keep labels from usage-to-payment ratios, then trained and benchmarked Logistic Regression, Random Forest, and Gradient Boosting classifiers, flagging 4,782 subscribers for upgrade (an estimated 743K+ EGP/month in recoverable revenue) and 32 for downgrade (an estimated 72K EGP/month in over-billing risk).
- Built a multilingual (English/Arabic/Franco-Arabic) customer support-ticket bot on a custom-generated dataset of ~3,000 synthetic tickets: fine-tuned MARBERTv2 to auto-route each ticket to the right department and flag customer sentiment, and fine-tuned mT5 to generate a short summary for the agent, deployed as a Streamlit dashboard for both single-ticket lookup and batch CSV processing.
- Built a Retrieval-Augmented Generation (RAG) customer-support chatbot for WE: scraped and chunked company web content, embedded it into a Qdrant vector store, and served bilingual (Arabic/English) answers grounded strictly in retrieved context via an LLM pipeline.
- Built Project Synapse, an internal Streamlit tool that turns the quota-exhaustion, bad-line-billing, and outage/fraud-detection models I built into a working product: a customer-facing portal showing each household's status in plain language, alongside a review console for testing and demoing the models to stakeholders.

AI Team Intern — Pulse by Solutions (formerly Giza Systems) *Aug 2026 – Sep 2026*

- Designed and containerized a reusable MLOps stack with Docker Compose (MLflow tracking server + Postgres backend, a Flask REST API, and an Airflow orchestration layer on its own network-joined compose file), so each service could be built, versioned, and restarted independently.
- Built a generic, model-agnostic training/inference pipeline (run_training_pipeline / run_inference_pipeline) wrapping Optuna hyperparameter search and MLflow experiment logging, so new classifiers are added via a small config script instead of duplicating tuning/registration code — demonstrated on both a numeric (image) classifier and a text (TF-IDF + Logistic Regression) classifier with a separately versioned vectorizer artifact.
- Implemented alias-based model promotion (models loaded by a 'production' alias, not a hardcoded version), and exposed the pipeline through a Flask API (/predict, /retrain, /jobs) with async retraining jobs, input validation, and structured error handling.
- Authored an Airflow DAG for scheduled model retraining across the registry, with health checks, registry-sync validation, per-model Airflow Variables for trial counts, and optional auto-promotion that only promotes a new model version when it beats the current production accuracy.
- Built a RAG assistant in n8n over GitLab's public Handbook: an ingestion workflow that fetches, cleans, and chunks source documents (tagged by source doc ID) into a Qdrant vector store, and a query workflow that retrieves top-k chunks, applies a relevance-score gate, and generates a grounded answer (or a refusal when the docs don't support one).
- Built a companion evaluation pipeline in n8n that loops a 15–20 question test set (including trap questions) through the RAG assistant, scores each answer with an LLM-judge on retrieval-correctness and answer-correctness against a structured JSON schema, logs results to a data table, and compiles a summary report of accuracy rates and failed questions with reasons.

Product Intern – Data Analyst — iSchool Egypt (Hybrid) *Aug 2026 – Sep 2026*

- Built an OCR pipeline to extract and structure fields from scanned documents into clean, analysis-ready tables, with validation checks and confidence-based flagging so low-quality extractions were routed for manual review instead of entering the dataset.
- Developed and evaluated an allocation model that distributes limited resources under business constraints, benchmarking it against a baseline and documenting assumptions so the product team could review and adjust the logic.
- Designed and built responsive web pages and internal interfaces from stakeholder requirements, focusing on usability and clear data presentation.
- Tested internal websites for functionality and usability, documenting issues and reporting them to the product team for resolution.

- Validated dashboards for accuracy by cross-checking figures against source data and verifying calculations, filters, and KPIs before they were shared with stakeholders.

PROJECTS

Exology – AI Finance Recorder & Accounting Dashboard [View Project]
Competitive Hackathon Project | n8n, Gemini 2.5 Flash, Google Sheets, Telegram Bot API

- Designed and deployed a fully autonomous AI finance recording system for SMBs via Telegram, accepting text, voice, and photo inputs in Arabic and English — eliminating manual bookkeeping.
- Engineered three n8n workflows: real-time transaction recording with Gemini 2.5 Flash AI extraction, a nightly 30-day financial forecasting engine, and a weekly AI CFO health score report.
- Implemented multi-modal AI pipeline (text → JSON, voice .ogg → transcription, receipt photo → structured data) with data validation rules and confidence scoring before writing to Google Sheets.
- Built automated cash drought detection with proactive Telegram alerts when projected 30-day balance turns negative, using rolling 90-day averages for forecast projections.

Olist Brazilian E-Commerce – BI Dashboard & Full Analytics Pipeline [View Project]
Academic Research Project | Python, Power BI, Scikit-learn, KMeans, Plotly, Pandas | 9 Datasets, 100K+ Orders

- Engineered an end-to-end preprocessing pipeline across 9 relational datasets (100K+ orders), resolving missing product dimensions via KNN imputation, deduplicating 1M+ geolocation records to a single ZIP-level median coordinate, and applying 99th-percentile winsorization to eliminate extreme price and freight outliers.
- Constructed a unified master analytical table by merging all 9 source tables on order and customer keys, engineering 10+ features including lead_time_days, delivery_delay_days, is_late flag, freight_ratio, and purchase hour/day-of-week signals to power both BI reporting and downstream segmentation.
- Segmented 96K+ customers into 4 value tiers (Champions, At-Risk, New, Dormant) using RFM and K-Means (k=4 selected via Elbow and Silhouette scoring), enabling targeted retention strategies estimated to recover 9% incremental revenue by shifting orders-per-customer from 1.01 to 1.10.
- Delivered a 4-page Power BI executive dashboard diagnosing a R$316K (26.76%) revenue shortfall across 5,673 orders, isolating a demand-side volume gap as the root cause despite 92.18% on-time delivery and 78.11% positive sentiment; backed by a 35-page strategic report with 9 prioritised recommendations including Boleto conversion optimisation projected to recover 2,000+ confirmed monthly orders.

FDA Drug Adverse Event Reports (2015–2026) | Classification Modeling [View Project]
Academic Research Project | Python, Scikit-learn, XGBoost, LightGBM, SMOTE, FAERS Database

- Conducted a 14-step EDA audit on 528,000 pharmacovigilance records (FAERS 2015-2026), identifying and removing 7 target-leakage columns and 9,371 Death-flagged text entries that would have inflated model accuracy by ~10% while producing a clinically useless classifier.
- Developed 7 domain-specific clinical risk features capturing compounding fatality patterns absent from raw columns, including is_elderly_polypharmacy, is_high_risk_route (8 FDA IV/intraspinal/intracardiac codes), is_polypharmacy_high (6+ concurrent drugs), and reaction_drug_ratio as a systemic toxicity proxy.
- Built a production-grade preprocessing pipeline resolving 72% missing patient weight via strategic exclusion, reducing 10,446-category text columns via frequency encoding, and applying SMOTE to rebalance a 10.3% minority class, expanding the training set from ~475K to ~860K synthetic-augmented records.
- Benchmarked 5 classifiers (Logistic Regression baseline, Decision Tree, Random Forest, XGBoost, LightGBM); selected Random Forest as the production model for best Precision/Recall/F1 balance on fatality prediction, evaluated on an untouched real-distribution test set of ~105K records.

Customer Churn Analysis & Retention Strategy [View Project]
Self-Initiated Project | Python, Pandas, NumPy, Seaborn, Matplotlib, EDA, KPI Design

- Cleaned, merged, and validated 3 multi-source datasets (customer records, support tickets, activity logs) across 1,000+ records, resolving inconsistent identifiers and missing engagement timestamps before any analysis.
- Identified an 18-point churn rate gap between free-tier (41%) and paid subscribers (23%) via EDA; engineered a custom 30-Day Zero-Activity Rate KPI that flagged at-risk users up to 30 days before cancellation, enabling proactive intervention windows.
- Translated findings into a strategic retention report recommending 3 high-impact initiatives including a free-to-paid conversion campaign targeting the 41% churning free-tier segment.

EDUCATION

Egyptian Russian University (ERU) *Oct 2023 – Jun 2027 (Expected)*
Bachelor of Science in Business Technology — Specialization: Business Analysis
Relevant Coursework: Data Analysis, Descriptive & Predictive Analytics, Business Intelligence, Statistics, Database Management, Data Warehouse

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Data Analyst Professional Certificate: DataCamp
- Introduction to Data Analytics: Coursera
- Soft Skills Professional Development: Academy of Scientific Research & Technology

ADDITIONAL INFORMATION

Languages: English (Professional Proficiency) | Arabic (Native)